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Time From Submission of Johns Hopkins University Trial Results to Posting on ClinicalTrials.gov

The US Food and Drug Administration Amendments Act of 2007 (FDAAA)[1] requires that applicable clinical trials (ACTs) submit results to ClinicalTrials.gov within 1 year of completion. ClinicalTrials.gov identifies trials that likely meet this definition as probable ACTs (pACTs). A complementary National Institutes of Health (NIH) policy requires that nonapplicable clinical trials (non-ACTs) funded by grants submitted to NIH from January 18, 2017, also submit results.[2]

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After investigators submit results, the National Library of Medicine reviews their quality and may request changes by sending "comments" to investigators. Whether the NIH comments on records, the NIH must post results within 30 calendar days of first submission; the FDAAA makes no exemption for quality review time, and the NIH has not implemented a process for posting records that do not pass review.[1,3] We evaluated the time between submitting and public posting of results.

Methods | We examined all records in the "JohnsHopkinsU" ClinicalTrials.gov account, with results submitted from January 1, 2017, to December 31, 2018. JohnsHopkinsU includes studies conducted by the Johns Hopkins University (JHU) School of Medicine and School of Nursing. Other parts of JHU, including Bloomberg School of Public Health and Sidney Kimmel Comprehensive Cancer Center, have separate ClinicalTrials.gov accounts. Records are categorized as ACTs, pACTs, or non-ACTs. We combined the ACTs with the pACTs in our analysis. We determined the number of submission cycles and the number of days between submission and public posting, including the number of days under review by NIH and JHU. The first submission cycle began the day investigators submitted results and concluded when results were either posted by NIH or the record was returned to JHU with comments. Records publicly posted by the NIH to ClinicalTrials.gov by January 7, 2019, were included in the analysis. We used information available publicly and only to JHU, with data and code available at https://osf.io/9paqz/. Analyses were performed using Stata, ver-

Table. Time From Submission of Johns Hopkins University Clinical Trial Results to Results Posted, 2017-2018

Variable	FDAAA Status, Mean (SD)	
	ACT/pACT[a]	Non-ACT
Records with posted results, No.	97	18
Submission cycles, No.		
Mean (SD)	2.26 (0.81)	2.22 (0.73)
Median (IQR)	2 (1)	2 (1)
Total time in review, d		
JHU	16.84 (22.64)	43.72 (56.97)
NIH	59.39 (26.07)	118.28 (98.52)
Total	76.23 (39.53)	162.00 (139.85)
No. of days to receive first comments[b]	32.68 (6.97)	63.19 (52.99)
Posted after first cycle, No./total No. (%)	12/97 (12)	2/18 (11)
Time to review in first cycle, d		
JHU	NA	NA
NIH	30.92 (10.06)	30.00 (1.41)
Not posted after first cycle, No.	85	16
Posted after second cycle, No./total No. (%)	57/85 (67)	11/16 (69)
Time to review in second cycle, d		
JHU	17.51 (22.58)	62.09 (66.08)
NIH	20.65 (12.76)	77.64 (93.72)
Not posted after second cycle, No.	28	5
Posted after third cycle, No./total No. (%)	20/28 (71)	4/5 (80)
Time to review in third cycle, d		
JHU	4.20 (4.80)	12.75 (22.19)
NIH	17.90 (12.74)	8.75 (13.10)
Not posted after third cycle, No.	8	1
Posted after fourth cycle, No./total No. (%)	7/8 (88)	1/1 (100)
Time to review in fourth cycle, d		
JHU	1 (0)[c]	1 (NA)[d]
NIH	16.14 (15.12)	0 (NA)
Not posted after fourth cycle, No.	1	0
Posted after fifth cycle, No./total No. (%)	1/1 (100)	0/0[e]
Time to review in fifth cycle, d		
JHU	33 (NA)	NA
NIH	32 (NA)	NA

Abbreviations: ACT, applicable clinical trial; FDAAA, US Food and Drug Administration Amendments Act of 2007; IQR, interquartile range; JHU, Johns Hopkins University; NA, not applicable; NIH, National Institutes of Health; Non-ACTs, nonapplicable clinical trials; pACTs, probable applicable clinical trials.

[a] The FDAAA requires results reporting for ACTs. ClinicalTrials.gov identifies trials that likely meet this definition as pACTs. We combined the ACTs with the pACTs in our analysis.

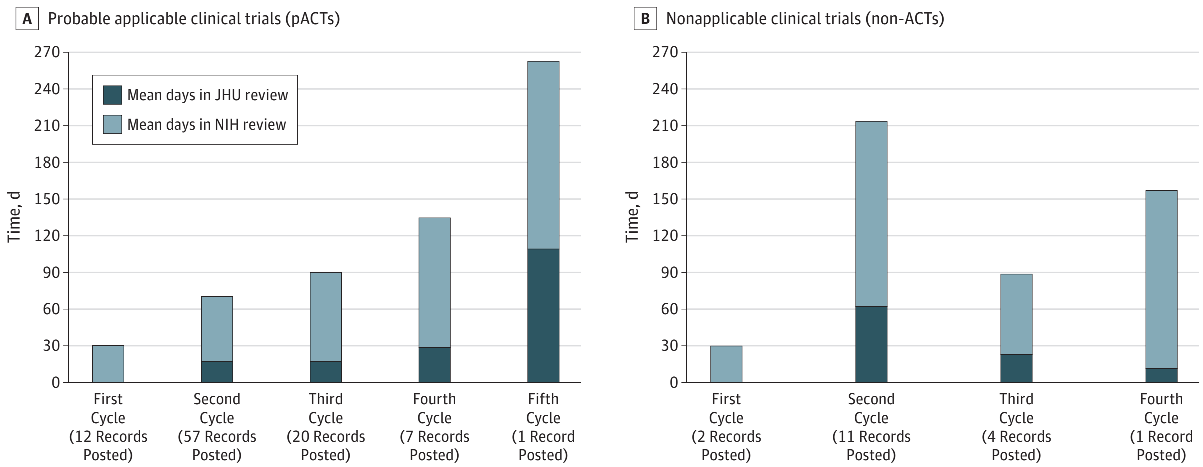
[b] If the NIH decides not to post a record, the NIH sends comments to the investigators, which ends one cycle and begins the next. In this analysis, we counted the day on which NIH comments were received as a JHU day.

[c] All 7 studies posted after the fourth submission cycle were modified and returned to NIH on the same day JHU received comments.

[d] Because there is 1 record, the SD is not applicable.

[e] All records were posted before this review cycle.

Figure. Days From Submission to Results Posted on ClinicalTrials.gov for Johns Hopkins University Clinical Trial Results Submitted 2017-2018



This figure shows the average number of days that records were in review by the National Institutes of Health (NIH) in light blue and Johns Hopkins University (JHU) in dark blue for records approved after a given number of submission cycles. The number below each bar is the number of records for which results were posted by NIH at the end of a given cycle (the Table shows the proportion of records that were posted after each number of cycles). Johns Hopkins University submitted results to ClinicalTrials.gov for 97 probable applicable clinical trials (pACTs) and 24 nonapplicable clinical trials (non-ACTs) (6 of these non-ACTS had not been posted by January 7, 2019, and are not included in the analysis). Thus, the first bar shows the average number of days in review for the 12 of 97 pACTs that were posted after first submission. Of pACTs submitted, 85 were submitted 2 or more times; the third bar shows the average number of days in review for the 57 of 85 pACTs that were posted after their second submission.

sion 15.1 (StataCorp) and were repeated stratified by record type, pACT (including ACTs) vs non-ACT, because non-ACT investigators and the NIH might give lower priority to voluntary submissions compared with required submissions. Because the analysis did not involve human subjects, there was no institutional review board submission, and patient consent was not obtained.

Results | Of 121 records submitted, we analyzed 115 records that posted results by January 7, 2019, including 97 of 115 pACTs (84%) and 18 of 115 non-ACTs (16%) (**Table**). Johns Hopkins University submitted records 1 to 5 times (mean [SD] = 2.25 [0.79]). The NIH sent first comments a mean (SD) of 32.68 (6.97) days after submission for pACTs and 63.19 (52.99) days after submission for non-ACTs. On average, pACTs were posted 76.23 (39.53) days after first submission; non-ACTs were posted after 162 (139.85) days. The NIH posted 7 of 97 pACTs (7%) within 30 days.

Discussion | The NIH took more than 30 days to comment on most clinical trial results submitted by JHU in 2017 and 2018. Furthermore, multiple submission cycles delayed posting of results (**Figure**). Although NIH quality review may ensure that clinical trial results are understandable and accurate, federal law requires that the NIH post results within 30 days without exception for quality review. For 93% of JHU records, the NIH exceeded the 30-day statutory limit for posting. The NIH might be able to post records faster as organizations that conduct clinical trials learn to identify errors and inconsistencies before submitting their results.[4] Several steps could be taken to ensure both that results are accurate and that they are posted in a timely manner. First, the NIH could summarize the types of comments that lead to multiple submissions, and the NIH could publish information about the duration of quality review, as we have done in this report. Sharing this information could help organizations that conduct clinical trials improve reporting. Second, academic organizations could support investigators by updating their policies and procedures and by hiring staff to support trial registration and reporting.[5] Administrators who interact with ClinicalTrials.gov regularly could support investigators who interact with ClinicalTrials.gov infrequently. Finally, the NIH could further improve automatic checking and reduce the time to review submitted records.

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Invited Commentary

The Culture of Trial Results Reporting at Academic Medical Centers

The ClinicalTrials.gov results database addresses problems with the published literature including publication bias, incomplete reporting of outcome measures and adverse events, and lack of transparency regarding the prespecified trial protocol. Submissions to ClinicalTrials.gov must contain a complete set of minimum results data elements. Entries undergo a quality-control review process based on criteria drawn from fundamental scientific principles, legal requirements, and experience operating the database.[1]

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Submissions that do not meet the review criteria are not posted but returned to submitters with specific comments. Common problems include inconsistent units of measure (eg, "time to relapse" with a unit of measure of "number of participants"), insufficient scale information (eg, "ABC scale" without a domain or range of possible values), inconsistencies within submissions (eg, outcome measure with 300 participants but 200 participants enrolled), and outcome measures lacking specificity (eg, efficacy).[2] Further, the most frequent problem for resubmissions is that previously identified problems remain unaddressed. Note that submitters may request one-on-one consultations with scientific review staff, including prior to submission or after receiving review comments.

Keyes et al[3] analyzed results first submitted by Johns Hopkins University (JHU) in 2017 and 2018. The authors reported that submissions required a median of 2 review cycles before meeting quality review criteria; some required up to 5 review cycles. Although they did not report the specific reasons that prevented posting of the JHU results submissions, these findings raise important questions.

First, why can investigators at a top academic medical center (AMC) not meet the ClinicalTrials.gov quality review criteria? While using a structured system with detailed requirements involves a steep learning curve, it has been more than 10 years since the US Food and Drug Administration Amendments Act of 2007 reporting requirements became effective. Johns Hopkins University should have plenty of experience. As with all AMCs, JHU has a legal, ethical, and scientific responsibility to ensure that sponsored trials are conducted and reported in accordance with relevant standards. It is not credible to assume that JHU investigators are unable to understand the data entry requirements, especially given the availability of one-on-one help from ClinicalTrials.gov. More likely is that they have not been sufficiently incentivized or supported in these efforts by their institution.

Johns Hopkins University is not unique. Generally, industry sponsors provide more support for results reporting and do better than AMCs in meeting the ClinicalTrials.gov requirements.[4] Although many AMCs have begun focusing on achieving timely reporting, they have not necessarily been concerned with submission quality. As illustrated in these data, low-quality submissions feed a cycle of resubmission that squanders resources for both the submitter and the National Library of Medicine, further increasing time to posting.

Second, what might be the impact of posting information about trial results that do not meet review criteria? As noted by Keyes et al,[3] the US Food and Drug Administration Amendments Act of 2007 requires that the National Institutes of Health (NIH) post results within 30 days of submission, even if they have clear errors. Presumably, lawmakers believed that this provision would enhance public access to important information. However, it is unclear how anybody could benefit from the availability of results that do not meet quality criteria and may be misleading or clearly erroneous.

In the ideal world, entries would almost always meet review criteria, and the NIH could post them within the 30-day time frame. In the real world, where most initial submissions have detectable flaws, the NIH must balance the competing goals of timeliness and information quality. The NIH has described their intended approach: posting all records within 30 days of submission, while implementing a method of alerting the public when a posted record has not met review criteria. Flagging records and noting the nature of the concern will be essential to alert users to a problematic entry to avoid inadvertent inclusion of flawed data in aggregate analyses and to facilitate the monitoring of reporting quality from different organizations. But flagging problems does not substitute for providing meaningful information.

The answer to these 2 questions invariably leads to others. Will AMCs and other sponsors be shamed into doing a better job once their results are posted with indications of their flaws? Will the US Food and Drug Administration and NIH be prompted to use their enforcement authorities, thus incentivizing trial sponsors to provide higher quality information? Or will the public come to distrust the ClinicalTrials.gov results database when so many posted entries display notices of errors?

Both the NIH and trial sponsors have a role to play in ensuring that accurate results are publicly available within the legally mandated time frame, but only the sponsor can provide the meaningful information. ClinicalTrials.gov continually works to improve the data entry system, increase real-time alerts to problematic entries, and increase staffing levels. These steps are likely to reduce frustration and burden of data entry but are unlikely to reduce most identified errors, which reflect difficulties understanding fundamental principles of trial methods or lack of access to the required data. These problems reflect deficiencies in the research enterprise that demand attention independent of ClinicalTrials.gov reporting. If the data in ClinicalTrials.gov cannot be trusted, then how can one trust journal publications that rely on the same information?

Although external pressure might incentivize improved reporting, in the end, the practices of individual investigators will continue to reflect the culture established by their leaders and the support of their organizations.

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Association Between Prescribed Opioids and Infections in Patients With Neutropenia and Cancer

In a recent article, Edelman and colleagues[1] identified an association between prescribed opioids and risk for community-acquired pneumonia among patients with and without HIV infection. This is of concern for immunocompromised patients receiving opioids for cancer pain. We used the prospective Cologne Cohort of Neutropenic Patients (NCT01821456) to confirm this observation in patients with chemotherapy-induced neutropenia.

Methods | The study was approved by the ethics committee of the University Hospital of Cologne, which waived written informed consent. We conducted a nested case-control study based on patients with cancer with an absolute neutrophil count of less than 500/μL treated between January 2008 and December 2018. A multivariable conditional logistic regression was performed resembling the analysis by Edelman et al[1] (excluding vaccines; Veterans Aging Cohort Study score; benzodiazepines, including antibacterial/antifungal prophylaxis; and Charlson comorbidity index[2]). Opioids were categorized as previously described,[1] adding buprenorphine to the nonimmunosuppressive groups[3] as well as piritramide and tilidine to the unknown group. Case study patients were inpatients with hospital-acquired infection (fever ≥38 °C, blood stream infection, pneumonia, or intensive care unit admission) and no baseline infection. Controls were matched 1:1 by age (aged within 5 years), sex, underlying disease, and time of admission (within 1 year) using the Gower coefficient. Sensitivity analyses were performed including (1) the effect of the opioid treatment duration, (2) the effect of preexisting outpatient opioid treatment, and (3) the association of opioid treatment with pneumonia only instead of the composite end point.

Results | We identified 2135 matched pairs with complete data sets, based on 3942 cases and 2949 controls in the cohort database. Opioids were administered in 371 of 2135 patient cases (17.4%) and 354 of 2135 patient controls (16.6%) (**Table**). No association of opioid administration was observed with the risk for infection (adjusted odds ratio [aOR], 0.84; 95% CI, 0.64-1.11 with immunosuppressive properties and aOR, 1.11; 95% CI, 0.86-1.43 without immunosuppressive properties) (Table). Patients with opioids from the unknown category, mostly meperidine (33 of 50 patient cases; 14 of 22 patient controls), showed an increased infection risk.

Sensitivity analysis revealed an association of longer opioid treatment with increased infection risk (aOR, 0.95; 95% CI, 0.92-0.98 with immunosuppressive properties and aOR, 0.98; 95% CI, 0.96-0.99 without immunosuppressive properties). Overall results were stable after excluding patients with preexisting opioid treatment from multivariable analysis (censored patient cases n = 143, patient controls n = 141; aOR, 0.97; 95% CI, 0.67-1.42 and aOR, 1.26; 95% CI, 0.91-1.76, respectively). There was also no relevant change when using only pneumonia as a clinical end point (patient cases n = 1072; aOR, 1.00; 95% CI, 0.75-1.34 with immunosuppressive properties and aOR, 1.10; 95% CI, 0.79-1.53 without immunosuppressive properties).

Discussion | In our analysis, we were not able to identify an increased risk for pneumonia or any infection by any type of opioid in severely immunocompromised patients with an overall infection risk of more than 50% during the observational